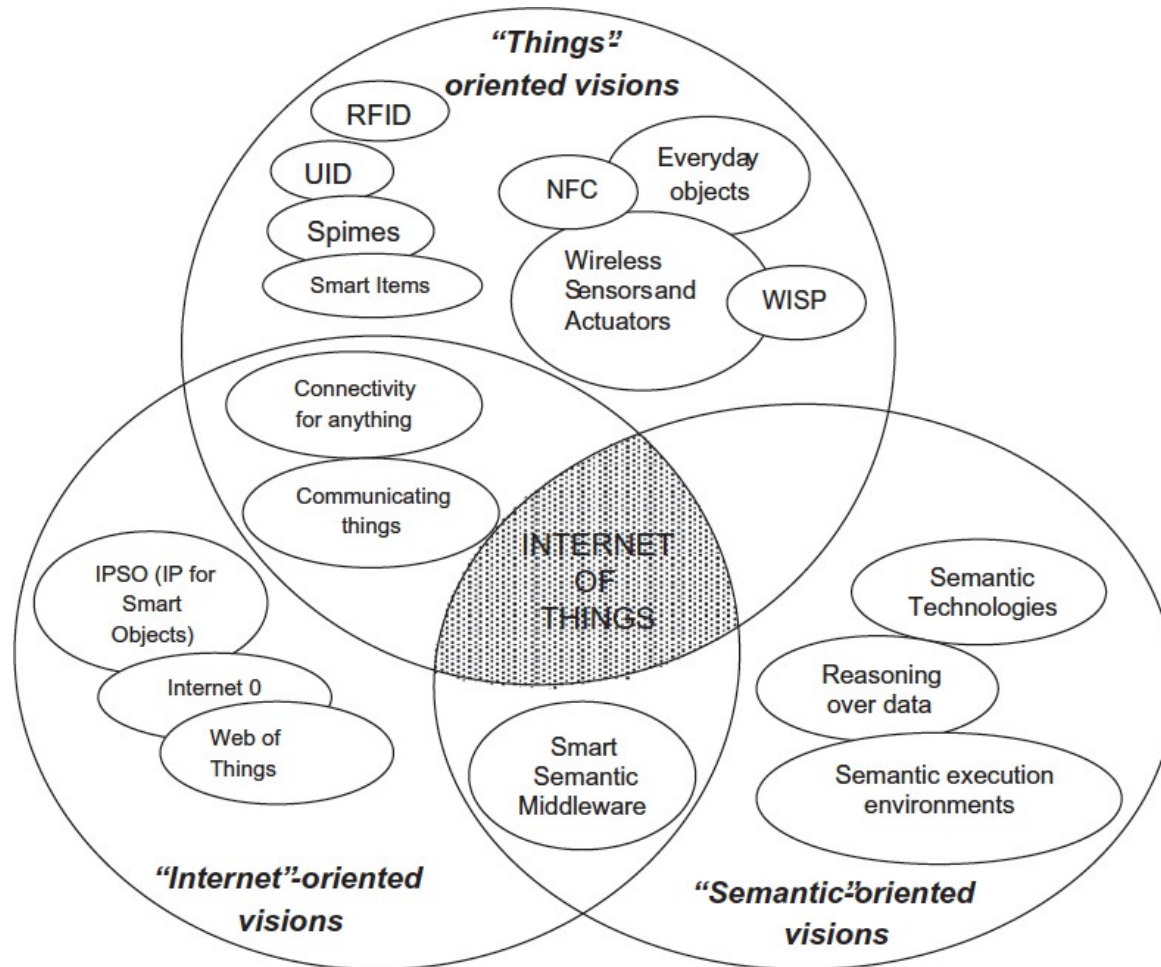


What is IoT?



- A phenomenon which connects a variety of ***things***
 - Everything that has the ability to communicate

Connection of Multiple Visions



IoT Definitions

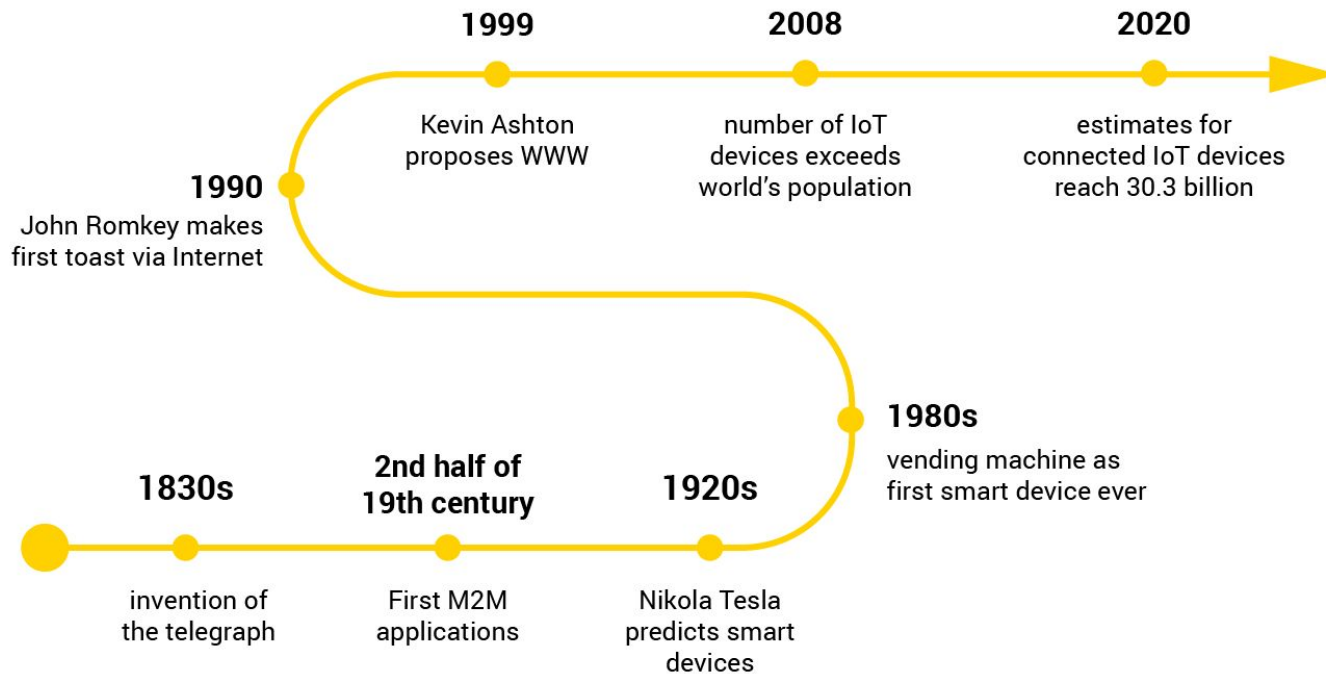
- The Internet of Things, also called The Internet of Objects, refers to a wireless network between objects, usually the network will be wireless and self-configuring, such as household appliances. **(Wikipedia)**
- The term "Internet of Things" has come to describe a number of technologies and research disciplines that enable the Internet to reach out into the real world of physical objects. **(IoT 2008)**
- “Things having identities and virtual personalities operating in smart spaces using intelligent interfaces to connect and communicate within social, environmental, and user contexts”. **(IoT in 2020)**

IoT Evolution



- Starts with only network and evolves into everything that can be connected with *a network*

History of IoT

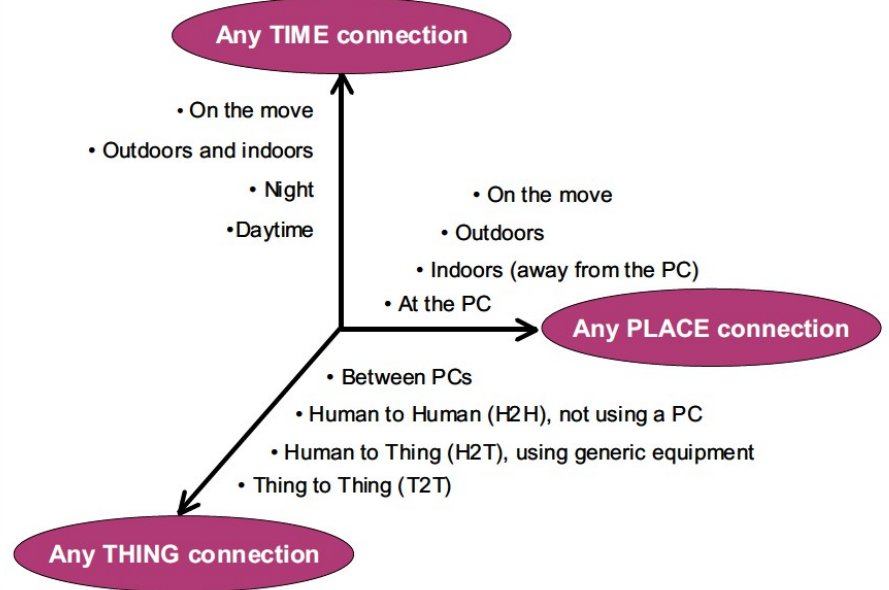


Any-X Point of View



Source: Perera et al. 2014

Figure 1 – A new dimension



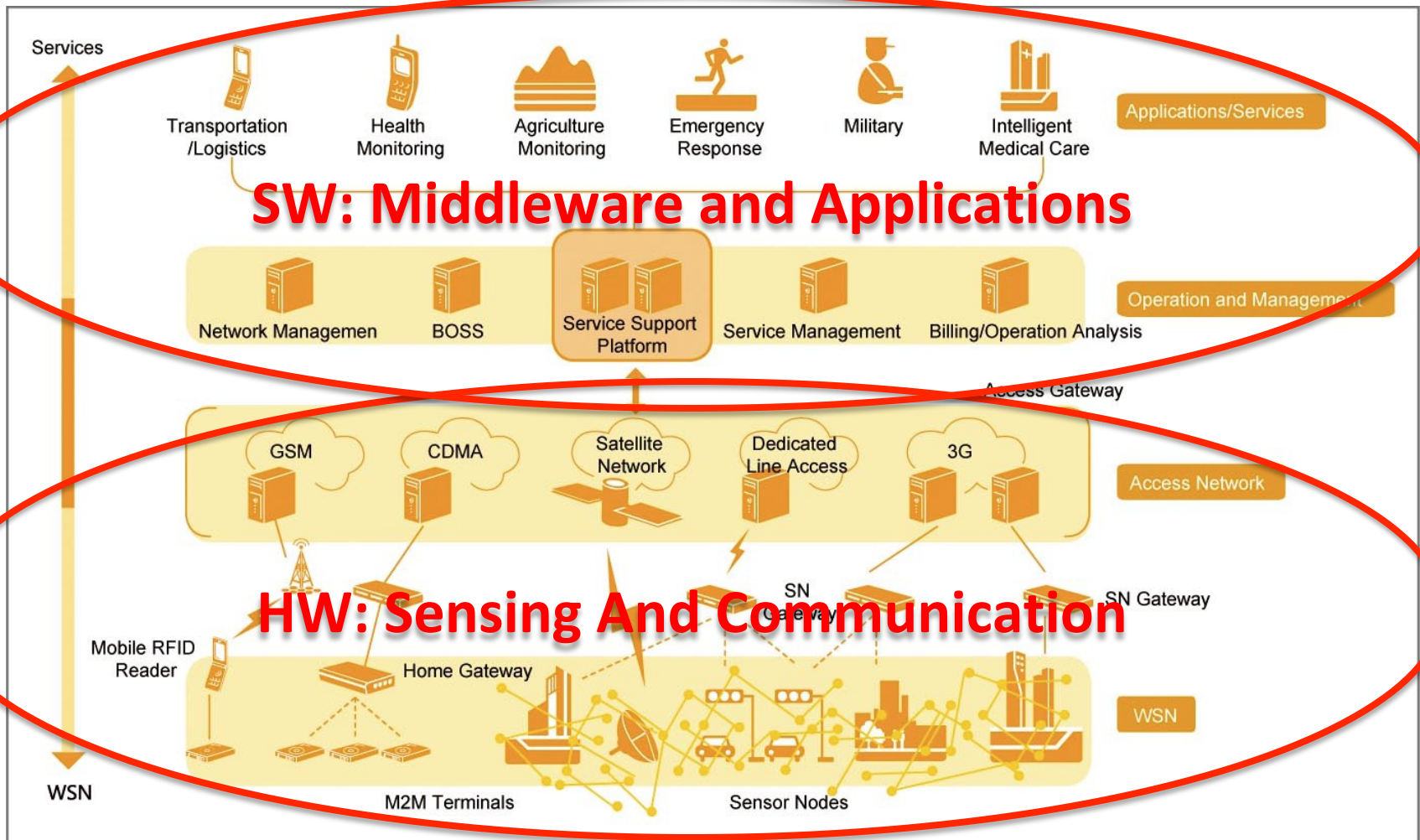
Source: ITU adapted from Nomura Research Institute

•The Internet of Things allows people and things to be connected Anytime, Anyplace, with Anything and Anyone, ideally using Any path/network and Any service.

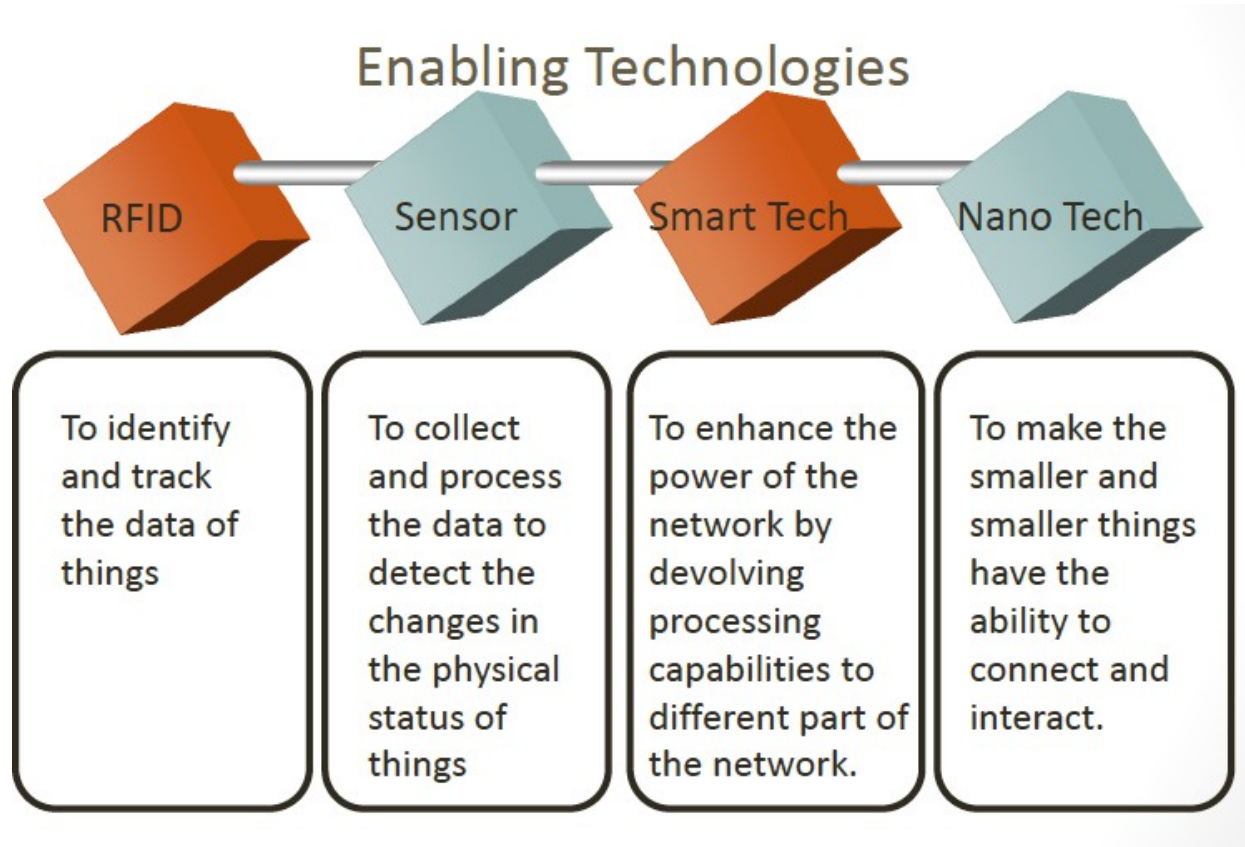
Characteristics of IoT

1. Intelligence
 - Knowledge extraction from the generated data
2. Architecture
 - A hybrid architecture supporting many others
3. Complex system
 - A diverse set of dynamically changing objects
4. Size considerations
 - Scalability
5. Time considerations
 - Billions of parallel and simultaneous events
6. Space considerations
 - Localization
7. Everything-as-a-service

IoT Layered Architecture



Networking and Communication



- RFID to smallest enabling technologies, such as chips, etc.
- Mobile platforms, such as sensors, phones, etc.

RFIDs

f size, weight, energy

f the radio takes us to a new era

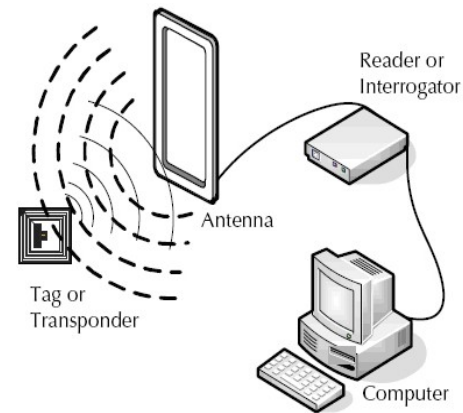
dios in almost all objects and thus, to
the above vision which leads to the

e readers and tags

chip attached to an antenna

e main, smallest

lects data



Wireless Technologies

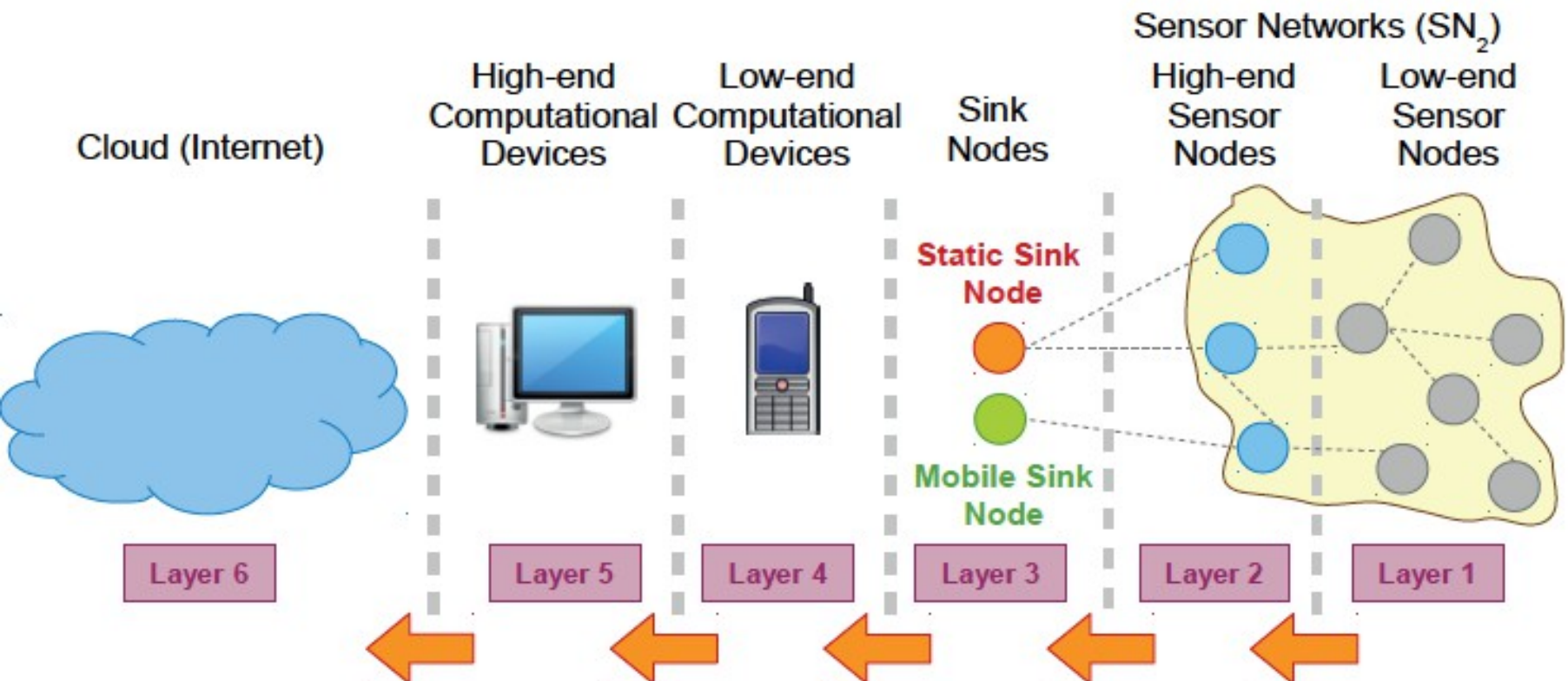
- Telecommunication systems
 - Initial/primary service: mobile voice telephony
 - Large coverage per access point (100s of meters – 10s of kilometers)
 - Low/moderate data rate (10s of kbit/s – 10s of Mbits/s)
 - Examples: GSM, UMTS, LTE
- WLAN
 - Initial service: Wireless Ethernet extension
 - Moderate coverage per access point (10s – 100s meters)
 - Moderate/high data rate (Mbits/s – 100s)
 - Examples: IEEE 802.11(a-g), Wimax

Wireless Technologies

- Short range:
 - Direct connection between devices – sensor networks
 - Typical low power usage
 - Examples: Bluetooth, Zigbee, Z-wave (house products)
- Other examples:
 - Satellite systems
 - Global coverage
 - Applications: audio/TV broadcast, positioning, personal communications
 - Broadcast systems
 - Satellite/terrestrial
 - Support for high speed mobiles
 - Fixed wireless access
 - Several technologies including DECT, WLAN, IEEE802.16, etc.

Sensor Networks (SNs)

- Consist of a certain number (which can be very high) of sensing nodes (generally wireless) communicating in a wireless multi hop fashion



Sensor Networks (SNs)

- SNs generally exist without IoT but IoT cannot exist without SNs
- SNs have been designed, developed, and used for specific application purposes
 - Environmental monitoring, agriculture, medical care, event detection etc.
- For IoT purposes, SNs need to have a middleware addressing these issues:
 - Abstraction support, data fusion, resource constraints, dynamic topology, application knowledge, programming paradigm, adaptability, scalability, security, and QoS support

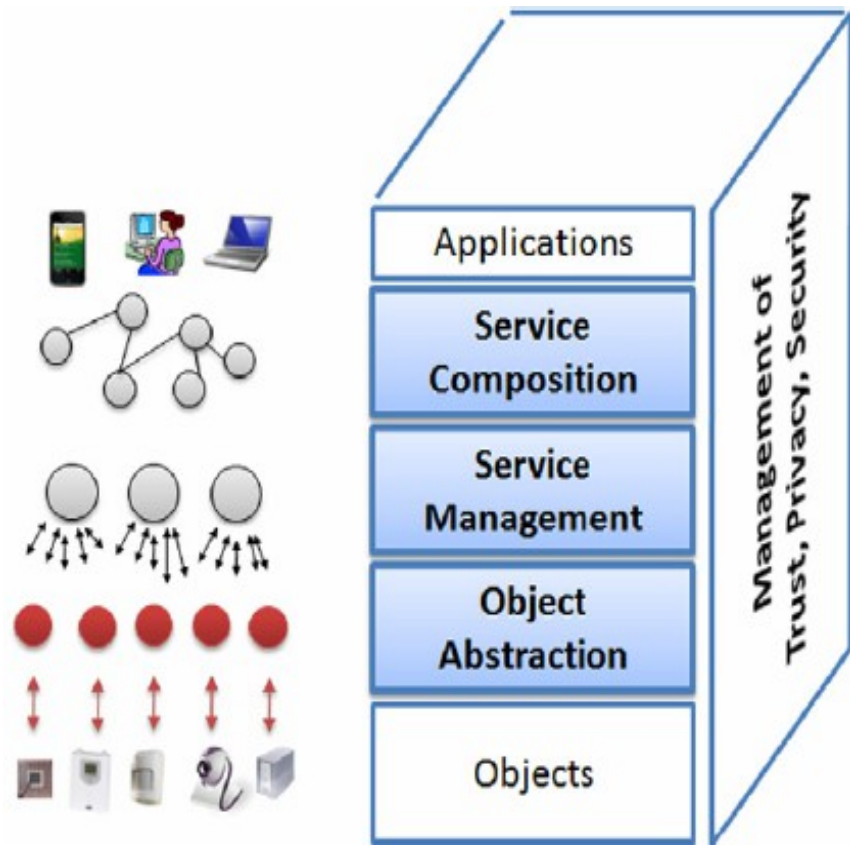
Example: Indoor Localization

- An indoor positioning system (IPS) is a solution to locate objects or people inside a building using radio waves, magnetic fields, acoustic signals, or other sensory information collected by mobile devices.
- For indoor localization:
 - Any wireless technology can be used for locating
 - GPS, WiFi, Bluetooth, RFID, Ultrawide band, Infrared, Visible light communication, Ultrasound

Middleware

- *Middleware is a software layer that stands between the networked operating system and the application and provides well known reusable solutions to frequently encountered problems like heterogeneity, interoperability, security, dependability [Issarny, 2008]*
- IoT requires stable and scalable middleware solutions to process the data coming from the networking layers

Service Oriented Architecture (SOA)



- Middleware solutions for IoT usually follow SOA approaches
- Allows SW/HW reuse
 - Doesn't impose specific technology
- A layered system model addressing previous issues
 - Abstraction, common services, composition

Other Middleware Examples

- Fosstrak Project
 - Data dissemination/aggregation/filtering/interpretation
 - Fault and configuration management, lookup and directory service, tag ID management, privacy
- Welbourne et al.
 - Tag an object/create-edit location info/combine events collected by antennas
- e-Sense Project
 - Middleware only collects data in a distributed fashion and transmits to actuators
- UbiSec&Sens Project
 - memory, etc.
– Focuses on security □ secure data collection, data store in

Open Problems and Challenges

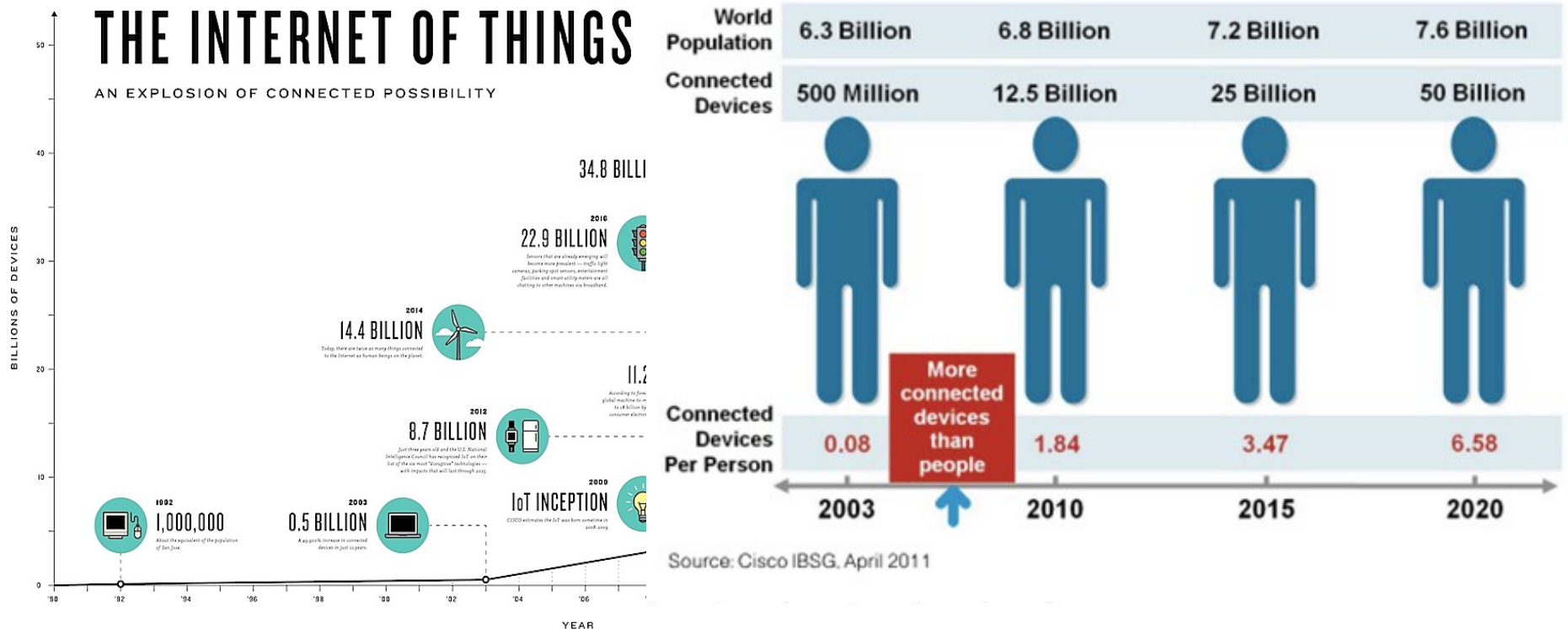
- Lack of standardization
- Scalability
 - Addressing issues
 - Understanding the big data
- Support for mobility
- Address acquisition
- New network traffic patterns to handle
- Security/Privacy issues

Standardization

- Several standardization efforts but not integrated in a comprehensive framework
- Open Interconnect Consortium: Atmell, Dell, Intel, Samsung and Wind River
- Industrial Internet Consortium: Intel, Cisco, GE, IBM
- AllSeen Alliance: Led by Qualcomm, many others

Standard	Objective	Status	Comm. range (m)	Data rate (kbps)	Unitary cost (\$)
<i>Standardization activities discussed in this section</i>					
EPCglobal	Integration of RFID technology into the electronic product code (EPC) framework, which allows for sharing of information related to products	Advanced	~1	~10 ²	~0.01
GRIFS	European Coordinated Action aimed at defining RFID standards supporting the transition from localized RFID applications to the <i>Internet of Things</i>	Ongoing	~1	~10 ²	~0.01
M2M	Definition of cost-effective solutions for machine-to-machine (M2M) communications, which should allow the related market to take off	Ongoing	N.S.	N.S.	N.S.
6LoWPAN	Integration of low-power IEEE 802.15.4 devices into IPv6 networks	Ongoing	10–100	~10 ²	~1
ROLL	Definition of routing protocols for heterogeneous low-power and lossy networks	Ongoing	N.S.	N.S.	N.S.
<i>Other relevant standardization activities</i>					
NFC	Definition of a set of protocols for low range and bidirectional communications	Advanced	~10 ⁻²	Up to 424	~0.1
Wireless Hart	Definition of protocols for self-organizing, self-healing and mesh architectures over IEEE 802.15.4 devices	Advanced	10–100	~10 ²	~1
ZigBee	Enabling reliable, cost-effective, low-power, wirelessly networked, monitoring and control products	Advanced	10–100	~10 ²	~1

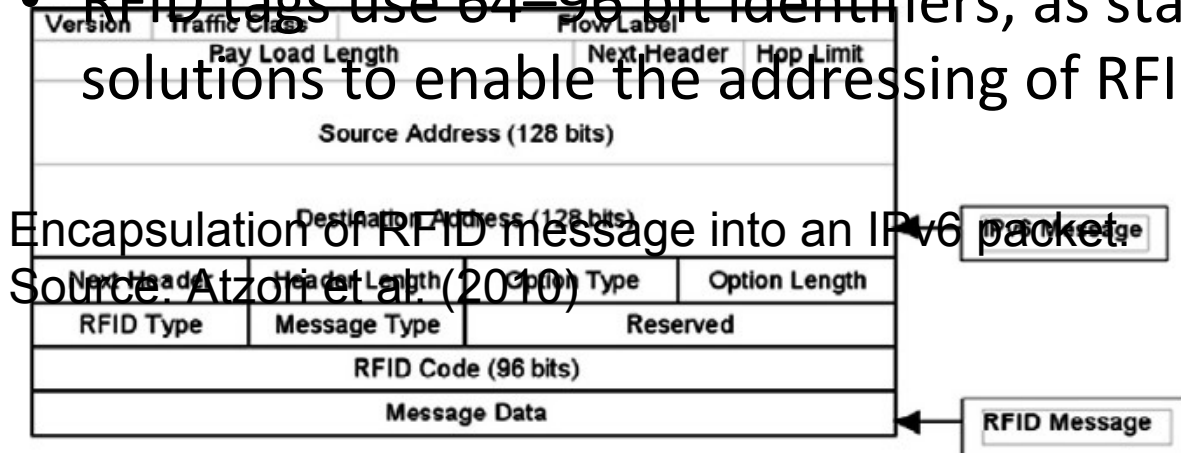
Scalability



- Number of devices increasing exponentially
 - How can they uniquely be tagged/named?
 - How can the data generated by these devices be managed?

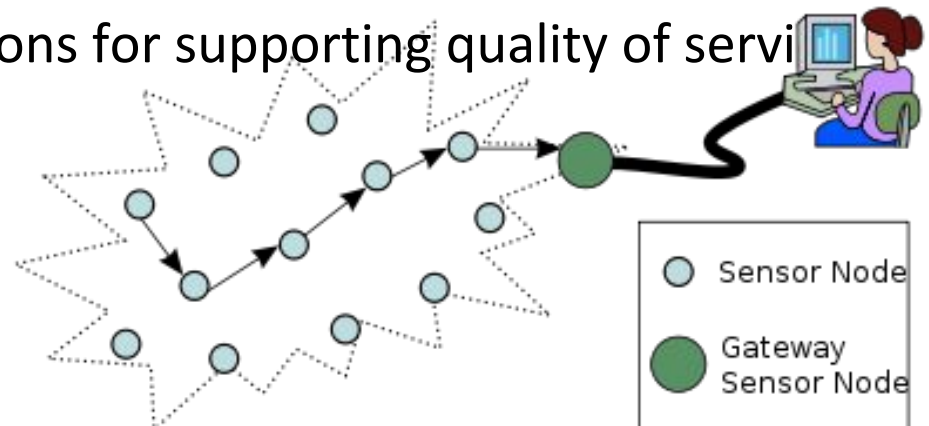
Addressing Issues

- Incredibly high number of nodes, each of which will produce content that should be retrievable by any authorized user
 - This requires effective addressing policies
 - IPv4 protocol may already reached its limit. Alternatives?
 - IPv6 addressing has been proposed for low-power wireless communication nodes within the 6LoWPAN context
- IPv6 addresses are expressed by means of 128 bits $\approx 10^{38}$ addresses, enough to identify objects worth to be addressed
- RFID tags use 64–96 bit identifiers, as standardized by EPCglobal, solutions to enable the addressing of RFID tags into IPv6 networks



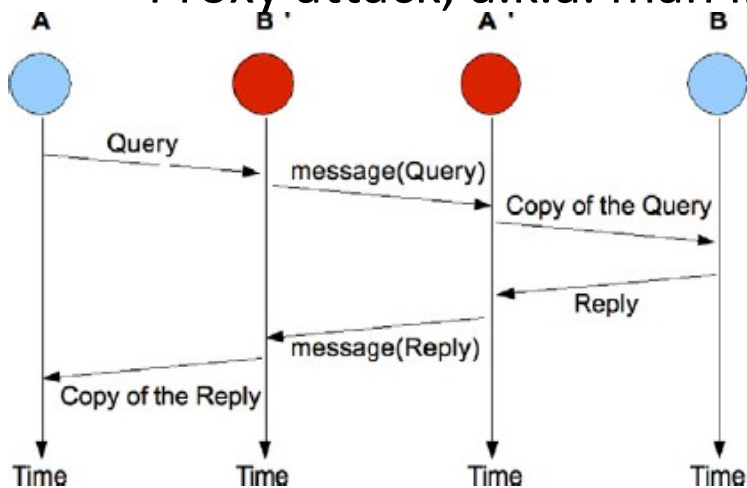
New Traffic to Handle

- The characteristics of the smart objects traffic in the IoT is still not known
 - Important $\square$ basis for the design of the network infrastructures and protocols
- Wireless sensor networks (WSNs) traffic characterization
 - Strongly depend on the application scenario
 - Problems arise when WSNs become part of the overall Internet
 - The Internet will be traversed by a large amount of data generated by sensor networks deployed for heterogeneous purposes $\square$ extremely different traffic characteristics
 - Required to devise good solutions for supporting quality of service



Security

- The components spend most of the time unattended
 - It is easy to physically attack them
- IoT components are characterized by low capabilities in terms of both energy and computing resources
 - They can't implement complex schemes supporting security
- Authentication problem
 - Proxy attack, a.k.a. man in the middle attack problem

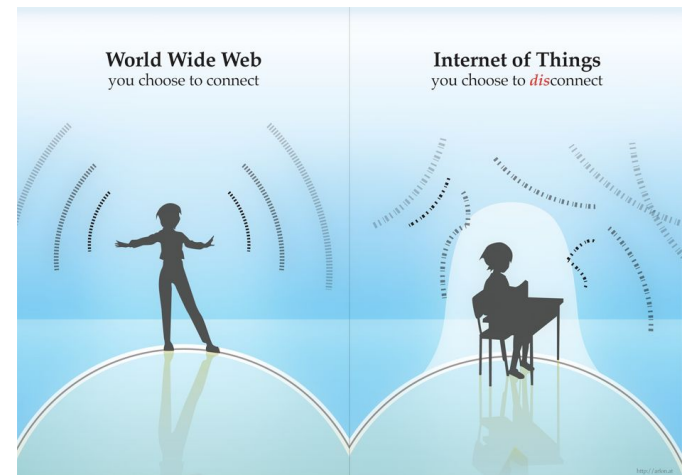


- Data integrity

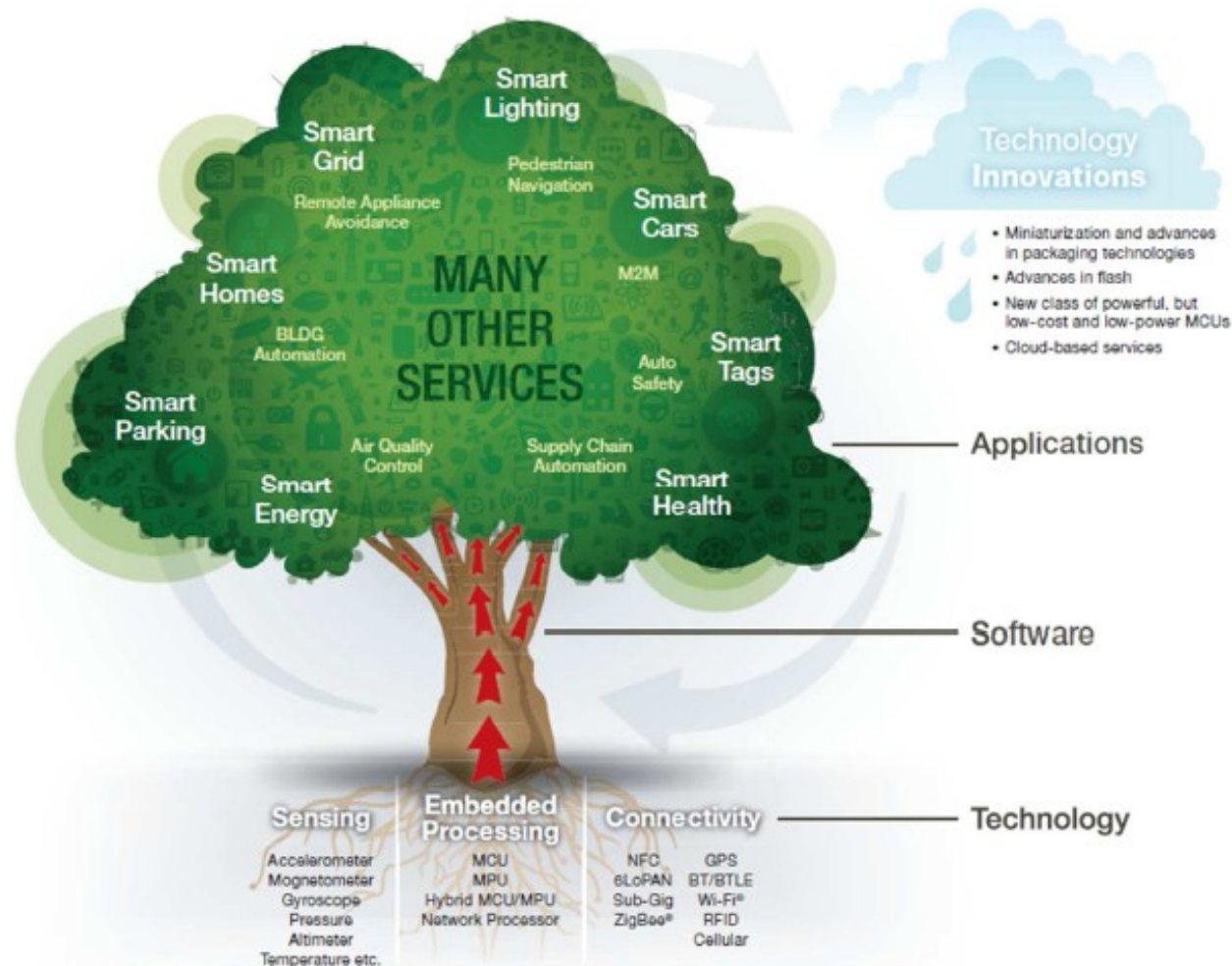
- Data should not be modified without the system detecting it
- Attacks on the node
 - Memory protection
- Attacks over the network

Privacy

- How is it different than traditional privacy?
 - Legislative issues
 - Ethics issues
- Easy for a person to get involved in IoT even if he/she does not know
- Data can be stored indefinitely
- Current solutions are not enough
 - Encryption, pseudo-noise signal, privacy broker

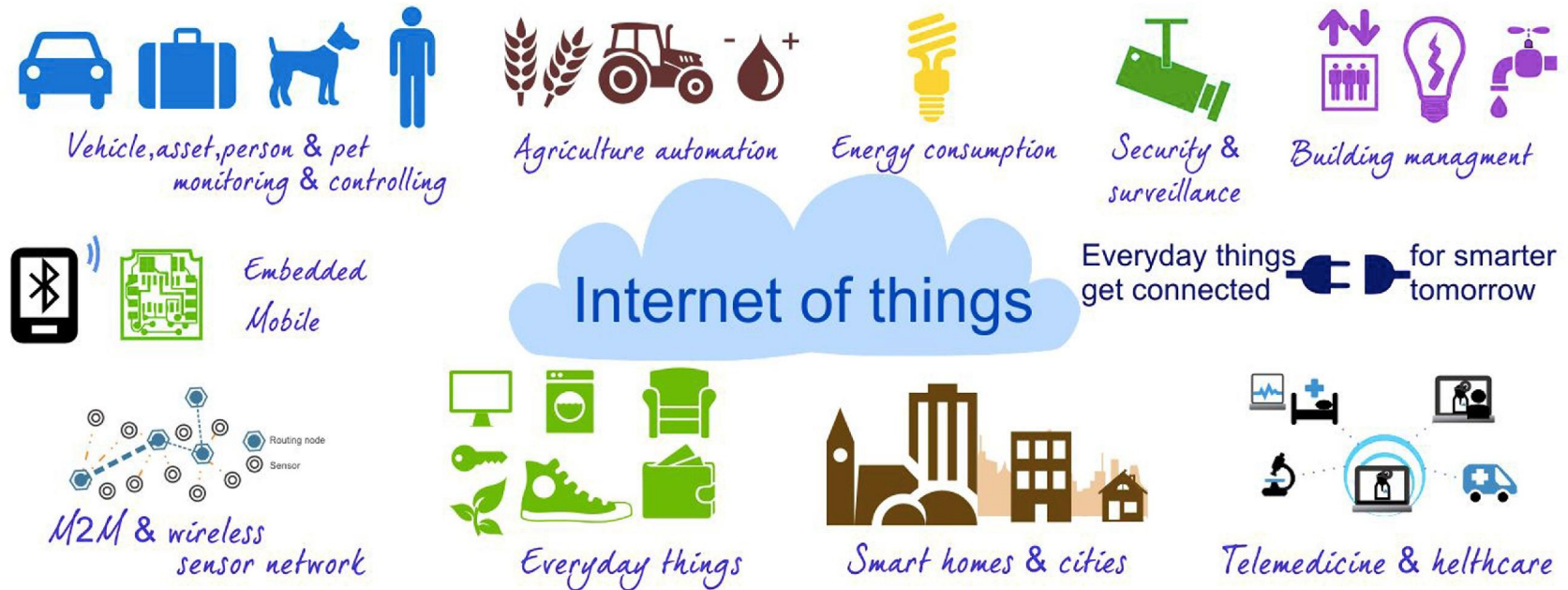


Putting it all together



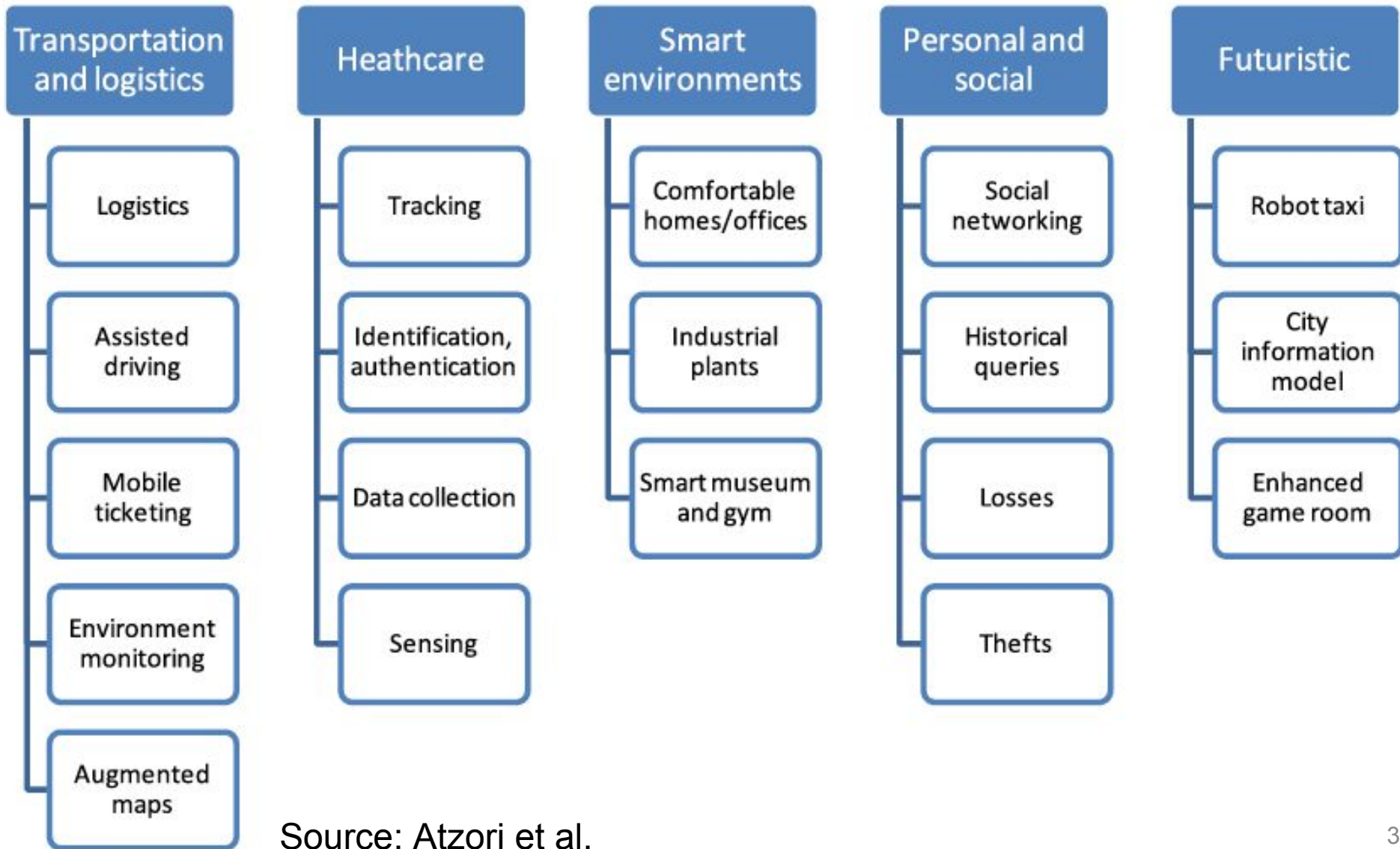
Source: "What the Internet of Things (IoT) Needs to Become a Reality," White Paper, by K. Karimi and G. Atkinson

Applications



- Several different domains
 - Transportation and logistics
 - Healthcare
 - Smart environment (home, office)
 - Personal and social domain
- Implications
 - Industry practices
 - IoT economics
 - Human behavior/habits

Application Domains and Scenarios



Source: Atzori et al.

Healthcare Applications

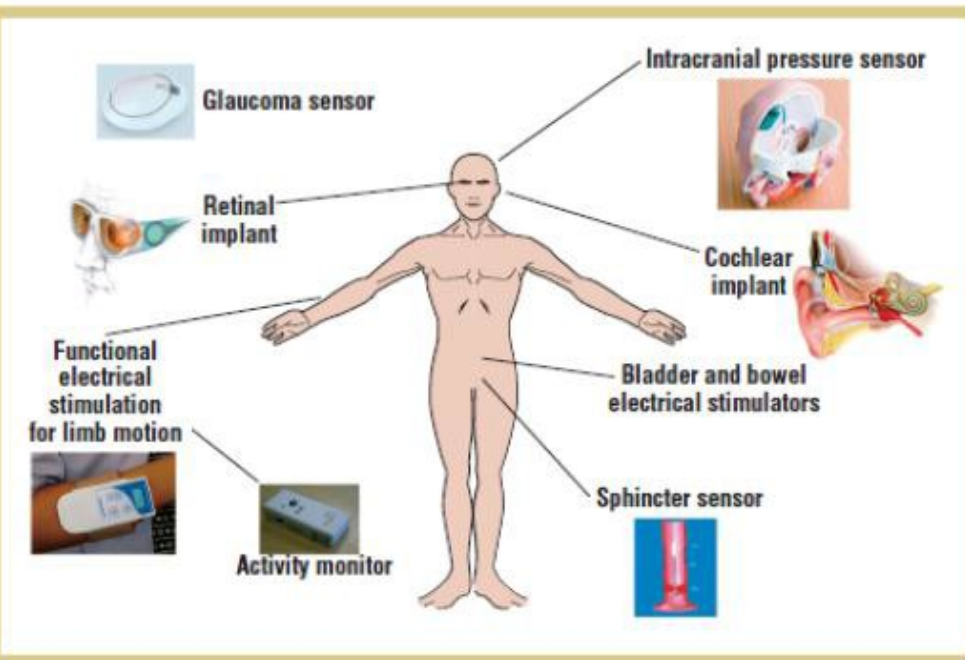


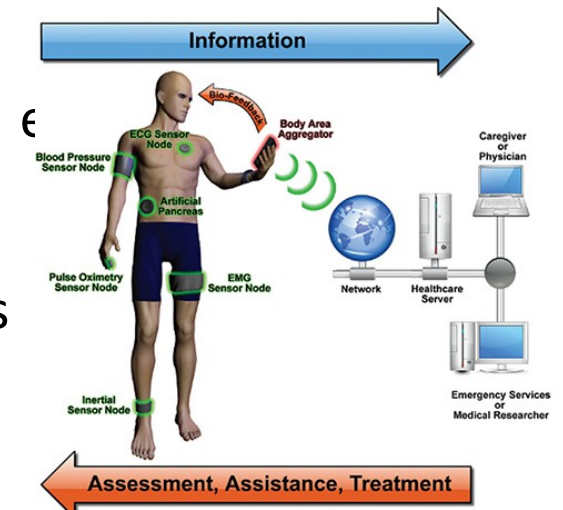
Figure 6. Fully implantable wireless sensor for the Intracranial pressure monitoring system.

- Various sensors for various conditions
- Example ICP sensor: Short or long term monitoring of pressure in the brain cavity
- Implanted in the brain cavity and senses the increase of pressure
- Sensor and associated electronics encapsulated in safe and biodegradable material
- External RF reader powers the unit and receives the signal
- Stability over 30 days so far

Healthcare Applications



- Other applications:
 - National Health Information Network
 - Electronic Patient Record
 - Home monitoring and control
 - Pulse oximeters, blood glucose monitors, infusion pumps, accelerometers
 - Bioinformatics
 - Gene/protein
 - Systems biology
 - Disease dynamics

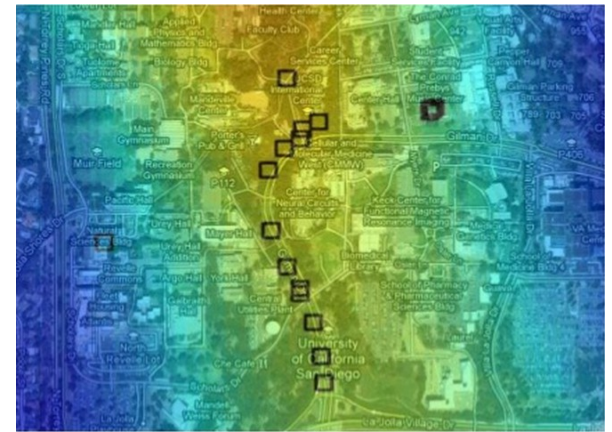


Environmental Application: CitiSense

- Air quality monitoring project in UCSD CSE

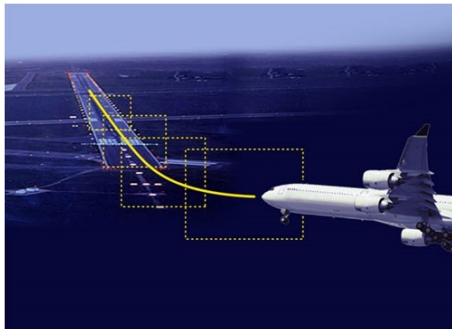


- Environmental application
- Electrochemical **sensors**, **microcontroller** for data collection and transmission to an **Android** app
- **Actuation**: air quality is immediately reported, as well as retransmitted to a backend for larger-scale analysis



Transportation Applications

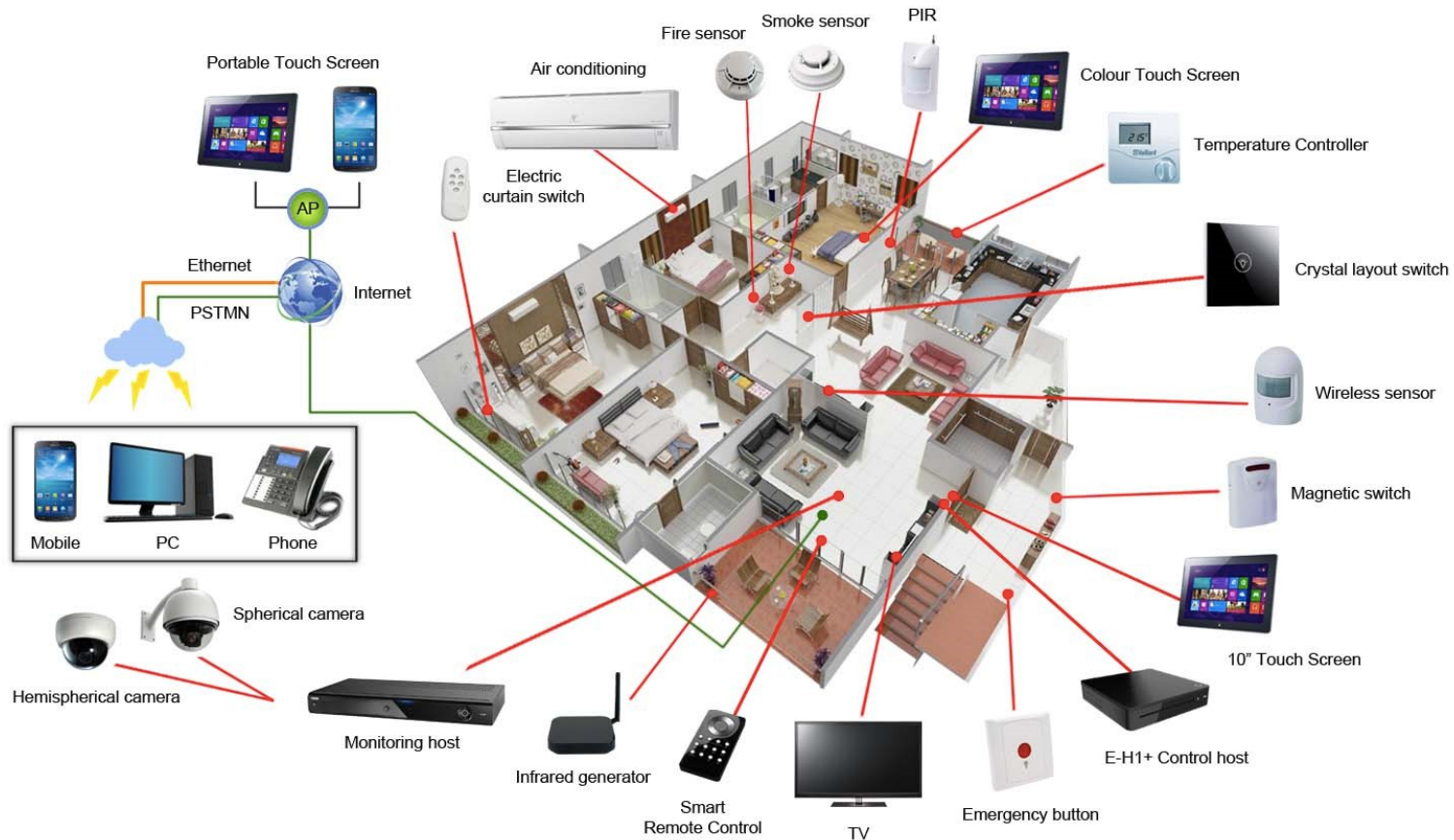
- **Vehicle control:** Airplanes, automobiles, autonomous vehicles
 - All kinds of sensors to provide accurate, redundant view of the world
 - Several processors in cars (Engine control, break system, airbag deployment system, windshield wiper, door locks, entertainment system, etc.)
 - Actuation is maintaining control of the vehicle
 - Very tight timing constraints and requirements enforced by the platforms



Example Transportation Scenarios

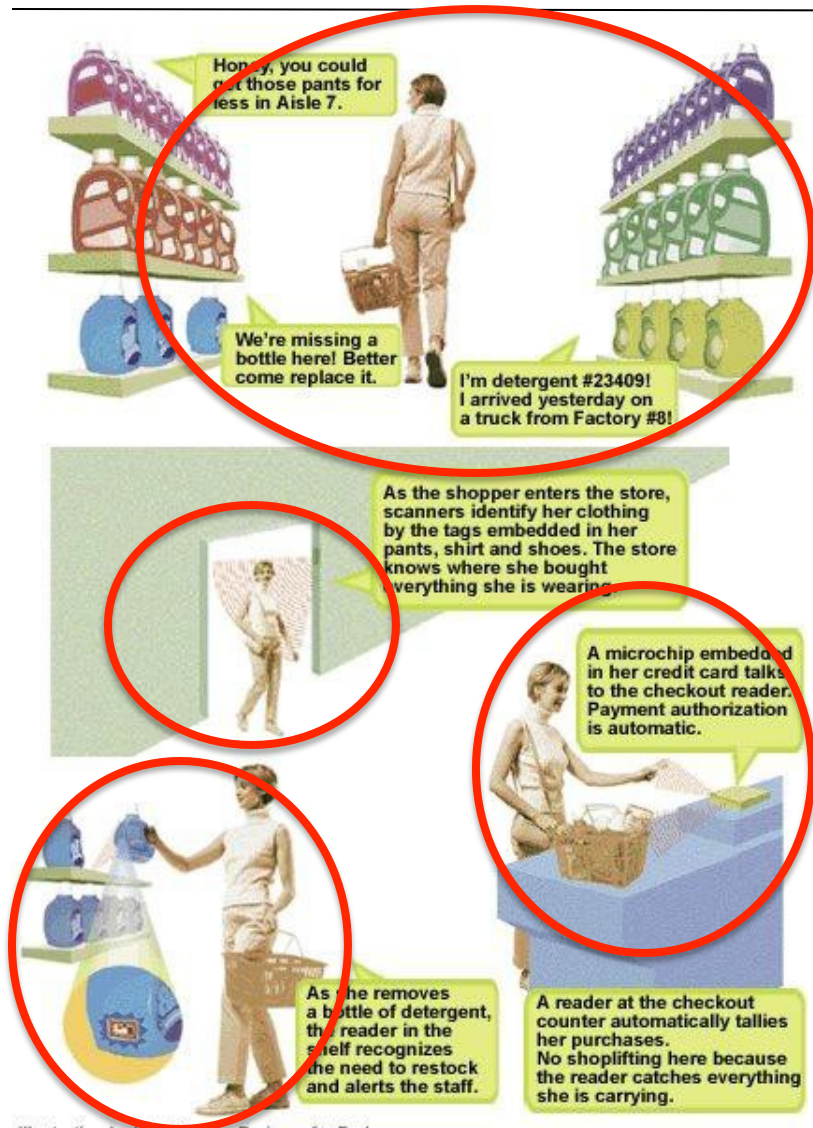
1. A network of sensors in a vehicle can interact with its surroundings to provide information
 - Local roads, weather and traffic conditions to the car driver
 - Adaptive drive systems to respond accordingly
2. Automatic activation of braking systems or speed control via fuel management systems.
 - Condition and event detection sensors can activate systems to maintain driver and passenger comfort and safety through the use of airbags and seatbelt pre-tensioning
3. Sensors for fatigue and mood monitoring based on driving conditions, driver behavior and facial indicators
 - Ensuring safe driving by activating warning systems or directly controlling the vehicle

Smart Home Applications



- Smart meters, heating/cooling, motion/temperature/lighting sensors, smart appliances, security, etc.

A Futuristic Application: Shopping



- When entering the doors, scanners will identify the tags on her clothing.
- When shopping in the market, the goods will introduce themselves.
- When paying for the goods, the microchip of the credit card will communicate with checkout reader.
- When moving the goods, the reader will tell the staff to put a new one.

An exciting future!

